

SIMATS

ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

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192311159RD

QUESTIONS

Q1

Student Registration System

10 marks

Q2

E-Commerce Cart System

10 marks

Q3

Library Management System

10 marks

Q4

Employee Data Processing

10 marks

Q5

Online Voting System

10 marks

5/5 answered

Q1 Student Registration System

10 marks

Suppose that a university system stores student data, where each student must be unique and can enroll in multiple courses.

(a) Enumerate three Collection interfaces suitable for this system.

(b) Design a Collection structure (diagram/representation) to store students and their enrolled courses.

(c) Write a Java program using HashSet and HashMap with Generics to:

- Add students
- Map students to courses
- Display data using Iterator

Your Code

```
1 import java.util.*;
2 public class StudentRegistrationSystem {
3     public static void main(String[] args) {
4         HashSet<String> students = new HashSet<>();
5         students.add("S101");
6         students.add("S102");
7         students.add("S103");
8         HashMap<String, ArrayList<String>> studentCourses = new HashMap<>();
9         studentCourses.put("S101", new ArrayList<>(Arrays.asList("Java", "Operating System")));
10        studentCourses.put("S102", new ArrayList<>(Arrays.asList("Java", "Operating System")));
11        studentCourses.put("S103", new ArrayList<>(Arrays.asList("Comp", "Operating System")));
12        System.out.println("Student Course Details");
13        System.out.println("-----");
14        Iterator<String> iterator = students.iterator();
15        while (iterator.hasNext()) {
16            String student = iterator.next();
17            System.out.println("Student ID : " + student);
18            System.out.println("Courses : " + studentCourses.get(student));
19            System.out.println();
20        }
21    }
22 }
```

Input

stdin input

Output

Student Course Details

Student ID : S102
Courses : [Java, Operating System]

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Q5 Online Voting System

10 marks

5/5 answered

Q5 Online Voting System

10 marks

```
3     public static void main(String[] args) {
4         HashSet<String> students = new HashSet<>();
5         students.add("S101");
6         students.add("S102");
7         students.add("S103");
8         HashMap<String, ArrayList<String>> studentCourses = new HashMap<>();
9         studentCourses.put("S101", new ArrayList<>(Arrays.asList("Java", "Operating System")));
10        studentCourses.put("S102", new ArrayList<>(Arrays.asList("Java", "Operating System")));
11        studentCourses.put("S103", new ArrayList<>(Arrays.asList("Comp", "Operating System")));
12        System.out.println("Student Course Details");
13        System.out.println("-----");
14        Iterator<String> iterator = students.iterator();
15        while (iterator.hasNext()) {
16            String student = iterator.next();
17            System.out.println("Student ID : " + student);
18            System.out.println("Courses : " + studentCourses.get(student));
19            System.out.println();
20        }
21    }
22 }
```

Output

Student Course Details

Student ID : S102
Courses : [Java, Operating System]

Run

Save

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